Minimally invasive suture training for resident, a pilot project: the MAESTRO project.

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Abstract

Despite the dominance of minimally invasive surgery (MIS), traditional "see one, do one, teach one" residency models often lack structured ex-vivo training. Competency Assessment Tools (CAT) are essential for objective performance evaluation in this field. To evaluate the effectiveness of a structured, 6-h intensive training protocol using the "Simsei laparoscopic trainer" in improving residents' skills in intracorporeal knot tying and continuous suturing. This prospective, observer-blinded study involved six surgical residents (PGY 1-5). Participants underwent a baseline test (intracorporeal knot and continuous suture), followed by 6 h of interval training over two months, and a final post-training test. Performances were recorded and evaluated by blinded raters using the validated Laparoscopic Suturing Competency Assessment Tool (LS-CAT). Statistical analysis was performed using paired t-tests. Baseline surveys revealed that while 83.3% of residents had used simulators, only 50% had them integrated into their formal curriculum. Post-training results showed significant improvements in overall LS-CAT scores, including instrument handling, tissue handling, and total error reduction. Specifically, significant progress was noted in almost all domains, except for needle passage and the first double throw (p=0.05 for both). A brief, structured ex-vivo curriculum significantly enhances laparoscopic suturing proficiency. These findings suggest that such programs should be integrated into core surgical residency curricula from the earliest stages of training. Further studies with larger cohorts and control groups are necessary to assess the transferability of these skills to the operating room.

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